MIL-F-50274 (MU)
12 November 1969

MILITARY SPECIFICATION

FUZES, GRENADE, H.E., M218, M219, M219E1 and XM224, LEAD CUP ASSEMBLY FOR LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Lead Cup Assembly for use in Grenade Fuzes H.E. M218, M219, M219E1 and XM224.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC: 1330

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8833562 - Lead Cup Assembly
8835999 - Box, Packing, Ammunition for Lead Cup Assembly, for Fuze Grenade, M219, M219E1, M218 or XM224
7548103 - Box, Set-Up, Packing, Ammunition, Outer for Small Explosive Components
7548104 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components

PUBLICATIONS

ARMY

ET-8833562 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

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3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with the applicable drawings and specifications.

3.2 Lead Cup Assembly.-The Lead Cup Assembly shall comply with all requirements specified on drawing 8833562, and with all requirements specified in applicable specifications.

3.3 Moisture Content of RDX.-The Moisture Content of the RDX composition at the time and site of loading shall not exceed 0.20 percent when determined as specified in 4.4.2.

3.4 Functioning Test.-The Lead Cup Assembly shall be capable of high order initiation by an M55 detonator across a minimum air gap of 0.060 inches. High order detonation will be evidenced by enlargement of hole in disc in excess of 3/16 inches in diameter when tested as specified in 4.4.1.

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base materials. All required markings shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5).

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services

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conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a sample of 340 assemblies to a Government approved facility designated by the contracting officer for testing in accordance with the provisions of 4.2.2 (See 6.1). All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.-Assemblies will be subjected by the Government to the following tests:

4.2.2.1 Functioning test (see 3.4).-Three hundred fifteen (315) Lead Cup Assemblies shall be subjected to this test. If

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six (6) or more assemblies fail to comply with applicable re-requirement, the first article sample shall be rejected. The test shall be performed as specified in 4.4.1.

4.2.2.2 Specific gravity of Lead Cup Charge.-Twenty-five (25) Loaded Lead Cup Assemblies shall be subjected to this test for conformance to requirements specified on applicable drawing.

4.2.3 Rejection.-In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchaged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Lead Cups of one lot interfix number from one manufacturer.
b. RDX from not more than one batch or blend of batches.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans

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for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Lead Cup Assembly (See Dwg. 8833562)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Length (max)	Gage	01001
102.	Flange diameter (max)	Gage	01002
103.	Small outside diameter (max)	Gage	01003
104.	Flange thickness	Gage	01004
105.	Explosive above or excessively below flush	Gage	01005
106.	Explosive on flange or exterior surface	Visual	01006
Minor:	AQL 1.00 Percent		
201.	Distortion	Visual	01007
202.	Evidence of poor workmanship (see 3.5)	Visual	01008

4.3.2.2 Box, Set-Up, Packing, Ammunition Inner For Small Explosive Components (See Dwg. 7548104)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	02001
102.	Number of lead cups incorrect	Visual	02002
Minor:	AQL 1.00 Percent		
201.	Marking misleading or unidentifiable	Visual	02004

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4.3.2.3 Box, Set-Up, Packing, Ammunition Outer For Small Explosive Components (See Dwg. 7548103)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Number of inner set-up boxes incorrect	Visual	03001
102.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal the box completely	Visual	03002
Minor:	AQL 1.50 Percent		
201.	Box torn, cut or punctured	Visual	03003
202.	Contents loose	Visual	03004
204.	Marking misleading or unidentifiable	Visual	03005

4.3.2.4 Box, Packing, Ammunition For Lead Cup Assembly For Fuze Grenade, M219, M219E1, M218 or XM224 (See Dwg. 8835999)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Hardware missing, broken or loose	Visual/Manual	04001
102.	Incorrect number of outer set-up boxes	Visual	04002
103.	Contents exposed	Visual	04003
104.	Strapping missing, broken or loose	Visual	04004
105.	Board broken or split	Visual	04005
Minor:	AQL 1.50 Percent		
201.	Hardware improperly engaged	Visual/Manual	04006
202.	Contents loose	Manual	04007
203.	Marking misleading or unidentifiable including color code	Visual	04008

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4.3.3 Testing

4.3.3.1 Composition (See Dwg. 8833562) Major Defect, Code No. 05001.-From the quantity of RDX used to produce a lot of Lead Cup Assemblies, at least one four (4)-ounce sample shall be selected for the determination of composition. Failure to comply with all of the specified drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Moisture content of RDX (See 3.3) Major Defect, Code No. 06001.-At the time and place of loading, a sample of approximately twenty (20) grams of unconsolidated material shall be selected from each lot and subjected to this test. If any sample fails to comply with the applicable requirement, the lot shall be rejected. The test shall be conducted as specified in 4.4.2.

4.3.3.3 Functioning Test (See 3.4) Major Defect, Code No. 07001.-Two hundred (200) Lead Cup Assemblies shall be selected from each lot for this test. If four (4) or more Lead Cup Assemblies fail to comply with the requirement specified in 3.4 the lot shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.4 Specific Gravity of Lead Cup Charge (See Dwg. 8833562) Major Defect, Code No. 08001.-A sample of five (5) Loaded Lead Cup assemblies shall be selected from each four (4) hours production and tested as specified in 4.4.3. If any sample fails to conform to the drawing requirements, the four (4) hours production shall be rejected.

4.3.3.5 Process functioning test (See Para. 3.4) Major Defect, Code No. 09001.-Twenty (20) Loaded Lead Cup Assemblies shall be selected (five (5) randomly selected per hour) from each four (4) hours production and subjected to this test. If any assembly fails to comply with the applicable requirement, the four (4) hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8833562 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

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4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures

4.4.1 Functioning test.-The Lead Cup Assembly shall be placed in a test fixture, over a steel Disc (.024 ± .003 thick), having a 1/16 hole centrally located, as shown on the equipment list. A weight of 1.00 ± 0.01 ounce shall be dropped from a height of one (1) foot on the firing pin. Observation shall be made for satisfactory functioning evidenced by the size of the hole in the disc. (See 3.4) In the event detonator does not fire, test shall be considered a "no test".

4.4.2 Moisture Content of RDX.-Transfer an accurately weighed portion of approximately 10.0 grams of the RDX composition from 4.3.3.2 to a tared weighing bottle with an outside ground cap. Place the bottle and contents with cap removed in an oven at 100 plus or minus 5 degrees C. for 40 to 50 minutes; cool in a desiccator, stopper and reweigh. Calculate the Loss in Weight of the bottle and contents as percent moisture in the sample.

4.4.3 Specific Gravity of Lead Cup Charge.-(Water Displacement Method).-The Loaded Lead Cup shall be weighed in air and the weight shall be recorded as A. The Loaded Lead Cup shall be weighed in water at a temperature of 25 C, and the weight shall be recorded as B. The volume of the Loaded Cup shall be determined by substracting B from A, (A-B) and the volume shall be recorded as C. The RDX composition shall be removed from the Lead Cup by washing the cup with acetone. The empty cup shall be weighed in air and the weight shall be recorded as D. The empty cup shall be weighed in water at 25 C, and the weight shall be recorded as E. The volume of the empty cup shall be determined by substracting E from D (D-E), and the volume shall be recorded as F. The volume of the RDX shall be determined by substracting the volume of the empty cup (F) from the volume of the Loaded Lead Cup (C), (C-F), and the volume shall be recorded as G, (C-F=G). The weight of the RDX compostion shall be determined by subtracting the weight of the empty Lead Cup in air (D) from the Loaded Lead Cup in air (A), and the weight shall be recorded as H, (A-D=H). The specific gravity shall be determined as follows:

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$$\text{Specific gravity} = \frac{H}{G}$$

G = Volume of RDX
H = Weight of RDX

NOTE: Analytical balance shall be capable of reading weights in milligrams with a sensitivity of 0.2 of a milligram.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The Lead Cup Assemblies shall be preserved and packaged in accordance with Dwgs. 7548103 and 7548104.

5.2 Packing

5.2.1 Level C.-The Lead Cup Assemblies shall be packed in accordance with Dwg. 8835999.

5.3 Marking.-Marking shall be in accordance with Dwg. 8835999, 7548103, 7548104 and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepared detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration

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instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Check Test for Deterioration.-If the total lapsed time between the original acceptance of any Lead Cup assembly lot and the assembly of that lot into the fuze exceeds two years, or if the Lead Cup assemblies have been subjected to adverse conditions, however brief, at any time since previous tests, the Lead Cup Assembly lot shall be subjected to and must satisfactorily pass the functional test, specified in 4.4.1, immediately before the Lead Cup Assembly lot is assembled into the Fuze. This test shall be performed by the contractor, on Lead Cup Assemblies, selected by the Government inspector, at the facility assembling Lead Cups into the fuze. (See 6.4.1 and 6.4.2).

6.4.1 Functioning Test.-Two hundred (200) Lead Cup Assemblies shall be selected and subjected to this test. If four (4) or more fail to comply with requirements specified in 3.4 the lot shall not be used in production.

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6.4.2 Cost of Check Test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the Lead Cup Assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the Lead Cup Assemblies or to the Contractor assembling the Lead Cup assemblies into the fuze, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the Lead Cup Assembly Lot was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

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